

Randomized and low-rank approximation

Sensitivity-dependent row sampling for ℓ_p embeddings when p exceeds two

Difficulty: challenging **Importance:** interesting to the community

Status: Open

Rating rationale: Challenging because ordinary sensitivities must control all directions with nearly linear total-sensitivity cost; community impact is instance-sensitive regression and subspace sketching.

Fix $p > 2$. For a full-column-rank matrix $A \in \mathbb{R}^{n \times d}$, with rows a_i^T , set

$$s_i = \sup_{x \neq 0} \frac{|a_i^T x|^p}{\|Ax\|_p^p}, \quad \mathcal{S} = \sum_{i=1}^n s_i.$$

For a scalar $\alpha > 0$, retain row i independently with probability

$$q_i = \min\{1, 1/n + s_i/\alpha\},$$

and scale each retained row by $q_i^{-1/p}$; let $\tilde{A}$ be the resulting matrix.

Do constants $C_p, c_p > 0$ exist such that, for every A and $0 < \varepsilon, \delta < 1/2$, one can choose α with

$$\sum_i q_i \leq C_p \varepsilon^{-2} (\mathcal{S} + d) \log^{c_p} \left(\frac{2nd}{\varepsilon \delta} \right),$$

while, with probability at least $1 - \delta$,

$$(1 - \varepsilon) \|Ax\|_p^p \leq \|\tilde{A}x\|_p^p \leq (1 + \varepsilon) \|Ax\|_p^p \quad \text{for all } x \in \mathbb{R}^d?$$

The size is the expected number of retained rows. The question concerns this independent sampling rule based on the ordinary ℓ_p sensitivities, not arbitrary row weights or augmented sensitivities.

This would exploit easy instances through total sensitivity rather than using the worst-case dimension-only bound. It applies directly to sketching ℓ_p regression and other computations on the column space of A . The $p > 2$ ordinary-sensitivity theorem of Woodruff–Yasuda has size $\tilde{O}_p(\varepsilon^{-2} \mathcal{S}^{2-2/p})$.

References

1. D. P. Woodruff and T. Yasuda, *Sharper Bounds for ℓ_p Sensitivity Sampling*. ICML 2023, arXiv:2306.00732v2. Definition 1.1 fixes the sampling model, Theorem 1.5 gives the $p > 2$ bound, and Section 3 explicitly conjectures $\tilde{O}(\epsilon^{-2}(\mathcal{S} + d))$. [Paper](#).
2. A. Munteanu and S. Omlor, *Optimal bounds for ℓ_p sensitivity sampling via ℓ_2 augmentation*. ICML 2024, PMLR 235, pp. 36769–36796. Question 1.2 and Section 1.1 distinguish the solved/obstructed $p \leq 2$ situation from the remaining $p > 2$ case. [Paper](#).
3. R. Chhaya, A. Dasgupta, D. Feldman, and S. Shit, *Deterministic Coreset for L_p Subspace*. arXiv:2601.00361 (2026). This claimed deterministic embedding result was withdrawn on May 15, 2026 because some proofs are incomplete; it is cited only to prevent a stale abstract from being mistaken for a resolution. [Paper](#).

Status check — 2026-09-08

Searched “ $p > 2$ sensitivity sampling conjecture 2025 2026”, “sensitivity sampling $\mathcal{S} + d$ ”, and the exact paper titles; checked the current arXiv records. No resolution of the stated $p > 2$ rule was found. Results using ℓ_2 augmentation for $p \leq 2$ do not settle it. The 2026 deterministic embedding claim was withdrawn (latest v3, May 15, 2026), so its abstract is not evidence that any open case has been resolved.

Audit — 2026-09-10

Rechecked [Woodruff–Yasuda, §3](#) and the [augmentation paper’s scope](#). The ordinary-sensitivity $p > 2$ target remains unresolved in these sources. Sensitivity-sampling follow-up searches found no resolution. Difficulty was raised to reflect the general uniform sampling obstruction.